

Joonsun Hwang

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EDUCATION

Carnegie Mellon University

PhD in Civil and Environmental Engineering, College of Engineering
Master of Science in Civil and Environmental Engineering, College of Engineering
GPA : 3.96 /4.00

Pittsburgh, PA, US
Jan 2025 – Present
Aug 2024 – May 2026

Relevant Courses : Deep Reinforcement Learning, Generative AI, Urban System Modeling, Human AI Interaction, Principles of Digital Twins, Introduction to Python Prototyping for Infrastructure Systems, Advanced Python for Infrastructure Systems, Data Management, Introduction to Machine Learning

Seoul National University

Bachelor of Architecture, College of Engineering
GPA : 3.70/4.00 (Cum Laude), Ranking : 9th out of 66 graduates

Seoul, S. Korea
Mar 2015 – Feb 2020

Relevant Courses : Building Technology, Building System, Building Structure Planning, Introduction to Construction Engineering and Management, Building Economics and Development, Design Computing, Mechanical and Electrical Systems for building, Advanced Building Environmental Technology, Architectural Design Studio, Architecture Environmental Planning, Building Materials, Building Codes and Regulations, Architecture and Urban Design, Architectural Design Theory, Architecture and Society, Korean Architecture, Architecture and Urbanism of Seoul

CAREER

HMHI Research Group (Advisor: Dr. Pingbo Tang)

Research Assistant

Pittsburgh, PA, US
Aug 2024 – Present

- Conduct research on human-AI collaboration in civil infrastructure and manufacturing.
- Participate in funded projects on human-machine collaboration in building materials, apparel, and cell therapy manufacturing.
- Lead design-engineering teams in course project design and interdisciplinary research with the School of Architecture focused on facilitating effective communication in AEC industry.

Capstone Asset Management Co., Ltd.

Associate

Seoul, S. Korea
Jun 2023 – Jul 2024

- Participated in setting up new fund of 225 bn KRW (including 100 bn KRW investment from KAMCO, quasi-governmental organization of Korea) to invest in non-performing financial loans of real estate projects.
- Managed real estate funds, including Travelodge Dongdaemun Hotel and Yongin AMCC Logistics.
- Managed the fund of investing in public offering stocks listed on KOSPI and KOSDAQ.

ROK CBR Defense Command, Republic of Korea Air Force (Specialty: Civil Engineer)

First Lieutenant

Seoul, S. Korea
Mar 2020 – May 2023

- Served as an officer in CBRN Defense Research Institute, investigating and managing soil contamination in the military, from July 2022 to May 2023.
- Served as a platoon leader of Korea CBRNE Special Force (1st platoon, 3rd company, 24th battalion) until June 2022.

Savills Korea Co., Ltd.

Intern

Seoul, S. Korea
Jan 2020 – Feb 2020

- Provided support in writing due diligence and market reports on logistics.
- Suggested some ideas about future trends of logistics and wrote a report.

DA Group Urban Design & Architecture Co., Ltd.

Intern

Seoul, S. Korea
Jul 2018 – Jul 2018

- Provided support in residential apartment design competition of LH on Goyang-Jichuk area.
- Researched residential trends of newlyweds, including variable floor plans and residential communities.

PUBLICATIONS

Park, S., McComb, C., Xiao, J., **Hwang, J.**, Tang, P. (2026) A Traceable AI-Agent Operation Framework Using Graph-Based Procedures in Nuclear Abnormal Scenario Simulation. *18th International Conference on Probabilistic Safety Assessment and Management*

Park, S., **Hwang, J.**, Jeon, K., Boring, L. R., Tang, P. (2026) BuildLink: Graph-Structured Procedural Representation with Incident Integration. *International Conference on Computing in Civil Engineering*

Hwang, J., Jeon, K., Tang, P. (2026) Human and AI Decision-Making under Multi-objective Scenarios in Demand-Driven Building Product Manufacturing. *International Conference on Computing in Civil Engineering*

Hwang, J., Jeon, K., Tang, P. (2025) Measuring Text-based Task Similarity to Enable Personalized Learning in Modular Construction: A Case Study in HVAC Product Manufacturing. *International Conference on Computing in Civil Engineering*

Xiong, R., Netser, Y., Tang, P., Li, B., **Hwang, J.** (2024) Socio-Technical Assessment of Generative AI Integration in Architecture, Engineering, and Construction (AEC) Workflows: An Empirical Study Using O*NET Occupational Taxonomy. *Advanced Engineering Informatics*

PROJECTS

Manufacturing Futures Institute (MFI) 2026

Pittsburgh, PA, US

Research Assistant

Sep 2026 – Present

- Project title: Digitizing Tacit Manufacturing Knowledge into Graph-Grounded, Human-Aligned Coaching Agents
- Conduct research to identify the OODA loop of roll former operators and develop explainable approaches for guiding operators using experts' tacit knowledge

Autodesk Research Residency Program

Boston, MA, US (Hybrid)

Researcher

May 2026 – Present

- Project title: AI-Enhanced Decision-Making Framework for Manufacturing Operators in Advanced Cutting System
- Study how operators leverage their expertise to configure laser cutting machines using machine-generated logs, eye-tracking data, and think-aloud protocols.

Manufacturing PA Innovation Program (Cell Therapy Manufacturing)

Pittsburgh, PA, US

Research Assistant

Oct 2025 – Present

- Lead monthly meeting with industry partner (Elevate Bio Inc.) and managing overall project schedule
- Provide research consulting on adaptive learning strategies to onboard new employees in cleanroom operations at the upcoming manufacturing facility.

Textile Good Innovation Foundry (TGIF) Manufacturing Process Simulation and Assessment

Pittsburgh, PA, US

Research Assistant

Jan 2025 – Aug 2026

- Led on-site data collection for sewing machine operations, including vibration and acoustic sensor measurements
- Developed a playbook and identified middleware approaches for utilizing raw sensor data to connect physical and digital layers, supporting demand surge scenario evaluation through cycle time analysis and needle status monitoring.

Manufacturing PA Innovation Program (Building Material Manufacturing)

Pittsburgh, PA, US

Research Assistant

Nov 2024 – Present

- Lead research on explainable digital twins for preventive maintenance and quality control using sensor measurements from stamping and pressing lines
- Conducted research on capturing operators' tacit knowledge in adjusting multiple configurations of the roll former setup process through reinforcement learning.(Ductmate Inc.)
- Conducted research comparing task similarity across different manufacturing lines using a knowledge graph and submitted a paper to 2025 I3CE (Title: Measuring Text-based Task Similarity to Enable Personalized Learning in Modular Construction: A Case Study in HVAC Product Manufacturing)

Empowering Civil Engineering with Generative AI: Opportunities, Risks, and Future Directions

Pittsburgh, PA, US

Research Assistant

May 2024 – Nov 2024

- Provided assistance with data collection, analysis and interpretation for the project, funded by Block Center 2023 Future of Work.
- Assisted in submitting a paper to Advanced Engineering Informatics, waiting for accept (Title: Transforming Architecture, Engineering, and Construction (AEC) Workflows with Generative AI: Opportunities, Risks, and Future Directions)

OTHER EXPERIENCE

Safety 21 Research Assistant Fellow

Fellow student

Pittsburgh, PA, US

Aug 2024 – Dec 2025

- Led a project to develop a web-based auto-scraping tool to search for recent research by transportation faculty members.
- Conducted summarization of transportation articles and creation of research recaps, and provided assistance with multiple Safety 21 annual and special events held in Pittsburgh and Washington, D.C.

School-life Mentoring Program (SMP) for Foreign Students.

Team Leader

Seoul, S. Korea

Mar 2017 – Aug 2017

- Provided advisory on school life and general life in Korea for foreign freshmen as a mentor.
- Planned and organized opening party, membership training and tour programs around Seoul as a team leader.

Daewoo Engineering & Construction Student Ambassador

Student Ambassador

Seoul, S. Korea

Sep 2016 – Dec 2016

- Participated in CSR (Corporate Social Responsibility) activities.
- Reformed facilities and interiors of the lounge for cleaners at Sejong Univ, local youth welfare center in Wolgye-dong, and the library lounge at Konkuk Univ.

AWARDS

The 10th Anniversary of LH's Founding Idea Competition

Oct 2019 – Nov 2019

- Idea contest for corporation's future business and policy, hosted by LH (Land and Housing Corporation, quasi-governmental organization of Korea)
- Suggested senior-friendly architectural measures to improve the environment of aged public rental apartments.

The Synergy Architectural Design & MD Competition

Jul 2019 – Aug 2019

- Idea contest to build a retail facility in Cheomdan District, Gwangju, hosted by developer Synergy Tower Co., Ltd.
- Proposed idea and provided renderings of festive shopping space utilizing skip floor and continuous hallway.

The 20th Anniversary of Real Estate 114's Founding Big Data Competition

Apr 2019 – May 2019

- Idea contest for better use of corporation's real estate statistics program REPS, hosted by Real Estate 114 Co., Ltd.
- Analyzed real estate market price through classifying the New-Town release districts, using QGIS program.

International Idea Competition for Urban Innovations of Unused Space in Seoul

Dec 2018 – Feb 2019

- Idea contest to revitalize one of the 12 underused urban spaces, hosted by the Seoul Metropolitan Government.
- Selected the underused space near Jeongnueng, provided building plans and renderings for retail and student residences.

2018 Koramco Real Estate Challenge

Sep 2018 – Oct 2018

- Idea contest for the best use of the specific site concerning environmental-friendly as a view of the developer, hosted by Koramco REITs Management and Trust Co., Ltd.
- Provided business plan and cash flow of housing for in 20s to 30s on gas station site near Nakseongdae station, Seoul.

SCHOLARSHIP

Eminence Scholarship from Seoul National University (full tuition)

Spring, Fall 2019, Fall 2018

Kwanak Corporation Scholarship (merit-based, full tuition)

Spring, Fall 2017

Seoul Hope Fellowship from Seoul Scholarship Foundation (merit-based)

Spring, Fall 2017

Merit-based Scholarship from Seoul National University

Spring, Fall 2016, Fall 2015

CERTIFICATES

Licensed Real Estate Agent (KR): Real estate broker license in Korea

2021

Certified Investment Manager (KR): License to manage collective investment schemes, trusts or investment vehicle

2019

Engineer Architecture (KR): National authorized qualification in building construction with engineering knowledge

2019

TEACHING EXPERIENCE

Mentoring

Jan 2026 – Present

- Luke Cogswell (Summer Undergraduate Research Apprenticeship, 2026 Summer)
- Vera Zhou (Summer Undergraduate Research Apprenticeship, 2026 Summer)
- Donna Ding (Summer Undergraduate Research Apprenticeship, 2026 Summer)
- Soo-Yeon Kim (Summer Undergraduate Research Apprenticeship, 2026 Summer)
- Jingxuan Duan (Independent Study, 2026 Spring)

TA/Grader

Aug 2024 – Present

- Project Management for Engineering and Construction (TA, 2026 Fall, The number of students: 33)
- Project Management for Engineering and Construction (TA, 2025 Fall, The number of students: 50)
- Infrastructure Management (TA, 2025 Spring, The number of students: 28)
- Sustainable Engineering Principles (Grader, 2024 Fall, The number of students: 39)

Personal Lessons

Jan 2015 –Nov 2019

- Taught 8 high school students math, physics and chemistry for about 1,500 hours.

SKILLS

Tools for Architecture and Civil Engineering

- Design tools: AutoCad, Rhino, Grasshopper, Sketch-up
- Other tools: Revit, QGIS

Tools for Data Process, Analysis and Machine Learning

- Programming languages: Python, C++, Javascript
- Database / other languages: SQL, Flux, HTML/CSS